1. Title

Placement Data Analytics System

2. Introduction

College placement data contains important information about student recruitment, salaries offered, and hiring companies. Analyzing this data helps institutions understand placement performance and identify hiring trends.

The Placement Data Analytics System is developed using Python to analyze placement records and generate useful insights such as highest salary, average salary, company-wise selection count, and department-wise placement rate. The results are visualized using bar charts to make the analysis easier to understand.

3. Objective

The main objectives of this project are:

- To analyze student placement data.
- To identify the highest salary offered.
- To calculate the average salary of placed students.
- To determine the number of students selected by each company.
- To compute the department-wise placement rate.
- To visualize placement statistics using bar charts.

4. Tools and Technologies Used

Tool	**Purpose**
Python	Programming language used for analysis
Pandas	Data processing and analysis
Matplotlib	Data visualization
VS Code	Code editor used to develop the project
Excel	Used to create the dataset

5. Dataset Description

The dataset contains information about students and their placement details.

Column	Description
Student_ID	Unique student identifier
Name	Student name
Department	Student department
Company	Company where the student got placed
Salary_LPA	Salary offered in Lakhs per annum
Placed	Indicates whether the student is placed or not

Student_ID	Name	Department	Company	Salary_LPA	Placed
101	Rahul	CSE	TCS	6	Yes
102	Priya	ECE	Infosys	5	Yes
103	Arjun	CSE	Amazon	18	Yes
104	Meena	ME	None	0	No
105	Kiran	CSE	Accenture	7	Yes
106	Riya	ECE	Wipro	4.5	Yes
107	Vikram	ME	None	0	No
108	Asha	CSE	Google	25	Yes
109	Deepak	CIVIL	None	0	No
110	Sneha	ECE	TCS	6	Yes

6.Code

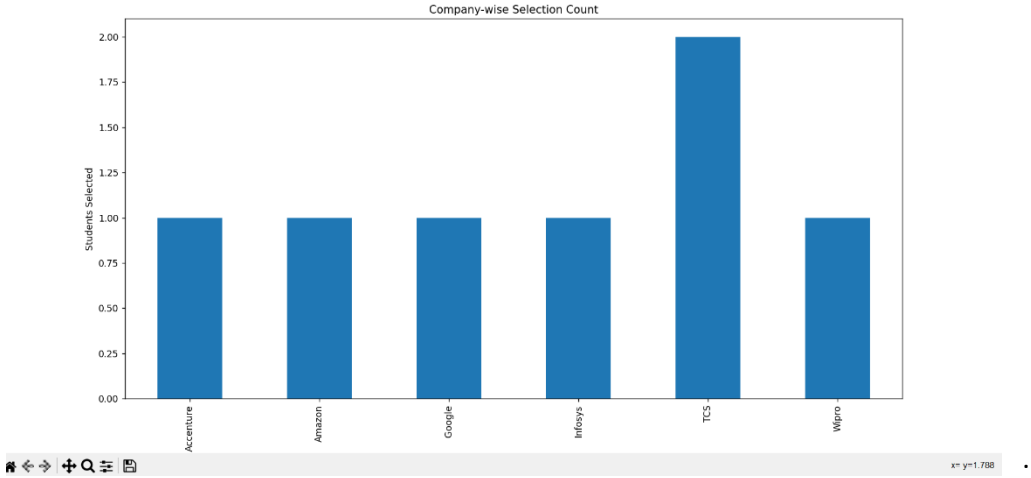
```
import pandas as pd
import matplotlib.pyplot as plt
# Load dataset
data = pd.read_csv("placement_data.csv")
print("\nPlacement Data\n")
print(data)
# Highest Salary
highest_salary = data["Salary_LPA"].max()
print("\nHighest Salary:", highest_salary, "LPA")
# Average Salary
average_salary = data["Salary_LPA"].mean()
print("Average Salary:", round(average_salary,2), "LPA")
# Company-wise Selection
company_count = data[data["Placed"]=="Yes"].groupby("Company").size()
print("\nCompany-wise Selection Count")
print(company_count)
# Department-wise Placement Rate
dept_total = data.groupby("Department").size()
dept_placed = data[data["Placed"]=="Yes"].groupby("Department").size()
placement_rate = (dept_placed / dept_total) * 100
print("\nDepartment-wise Placement Rate")
print(placement_rate)
# Bar Chart Company Selection
company_count.plot(kind="bar", title="Company-wise Selection Count")
plt.xlabel("Company")
plt.ylabel("Students Selected")
plt.show()
# Bar Chart Department Placement
placement_rate.plot(kind="bar", title="Department-wise Placement Rate")
plt.xlabel("Department")  plt.ylabel("Placement Percentage")  plt.show()
```

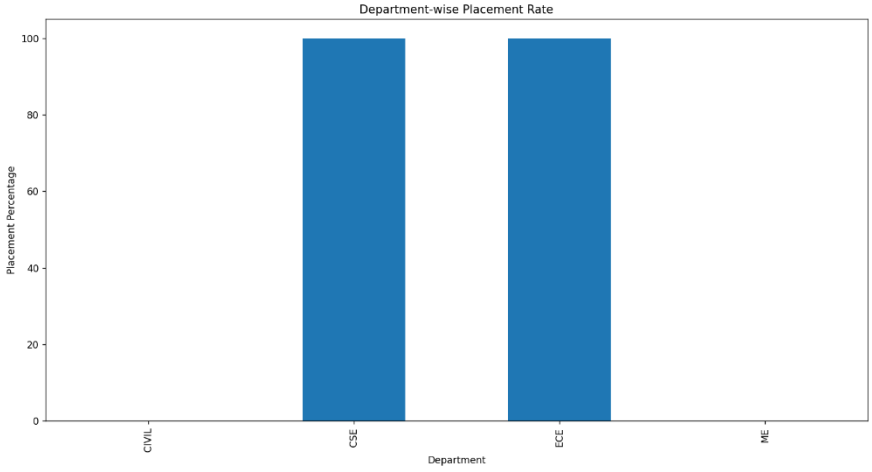
7. Results

```
Highest Salary: 25.0 LPA
Average Salary: 7.15 LPA

Company-wise Selection Count
Company
Accenture    1
Amazon       1
Google       1
Infosys      1
TCS          2
Wipro        1
dtype: int64

Department-wise Placement Rate
Department
CIVIL      NaN
CSE      100.0
ECE      100.0
ME         NaN
dtype: float64
```





Code zip file

C:\Users\samru\Desktop\Placement data analysis.zip

8. Conclusion

The Placement Data Analytics System successfully analyzes placement records and provides useful insights regarding salaries, hiring companies, and department performance. The use of Python and visualization libraries helps present the data in a clear and meaningful way. This system can assist institutions in understanding placement trends and improving recruitment strategies.